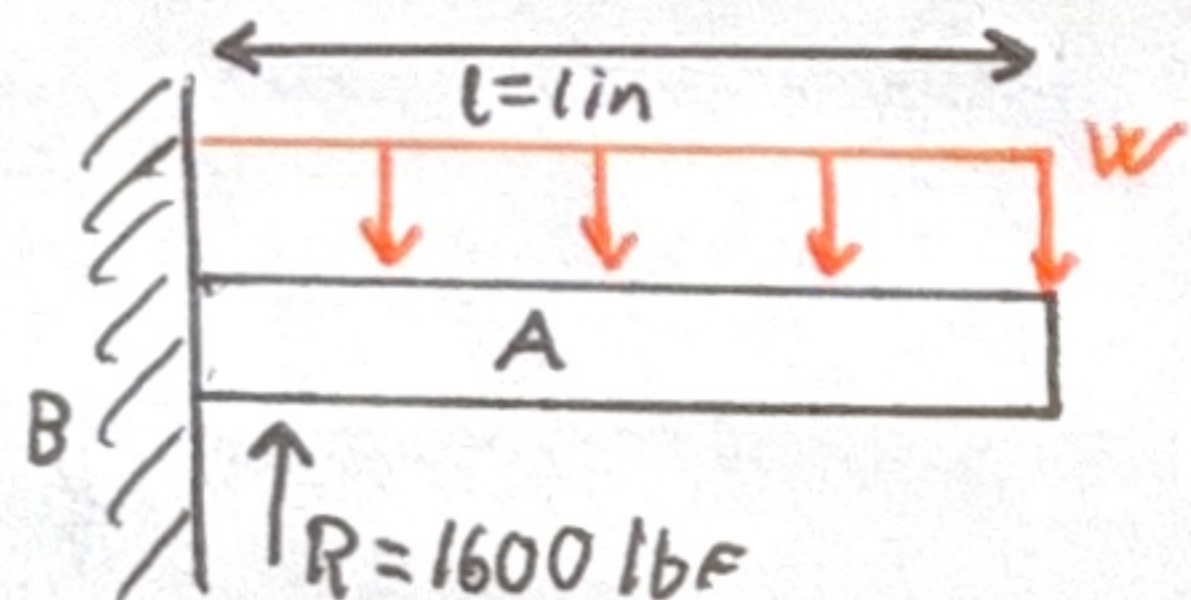


Feature A

Known:

$$W = 2F = 1600 \text{ lbf}, \quad l = 1 \text{ in}, \quad n = 4, \quad S_y = 39885 \text{ psi}$$

$$E = 10007604 \text{ psi}, \quad \delta_{\max} = 0.005 \text{ in}$$



$$\sum F_y = 0 \quad R - W = 0 \Rightarrow R = 1600 \text{ lbf}$$

$$\sum M = 0 \Rightarrow M - w\left(\frac{l}{2}\right) = 0 \quad M = 1600(0.5) = 800 \text{ in}\cdot\text{lbf}$$

$$\hookrightarrow M = 800 \text{ in}\cdot\text{lbf}$$

$$\sigma_{\text{allow}} = \frac{S_y}{n} = \frac{39885}{4} = 9971 \text{ psi}$$

$$\sigma = \frac{M}{Z} = \frac{Wl/2}{\pi r^3/4} = \frac{S_y}{n}$$

$$Z = \frac{nWl}{2S_y}, \quad r = \sqrt{\frac{4Z}{\pi}}$$

$$Z = \frac{4(1600)(1)}{2(39885)} = 0.0802 \text{ in}^3, \quad r = \sqrt[3]{\frac{4(0.0802)}{\pi}} = 0.4675 \text{ in} \Rightarrow \boxed{d = 0.935 \text{ in}}$$

$$\delta = \frac{Wl^3}{8EI} \Rightarrow I = \frac{Wl^3}{8E\delta_{\max}} = \frac{1600(1)^3}{8(10007604)(0.005)} = 0.003997 \text{ in}^4$$

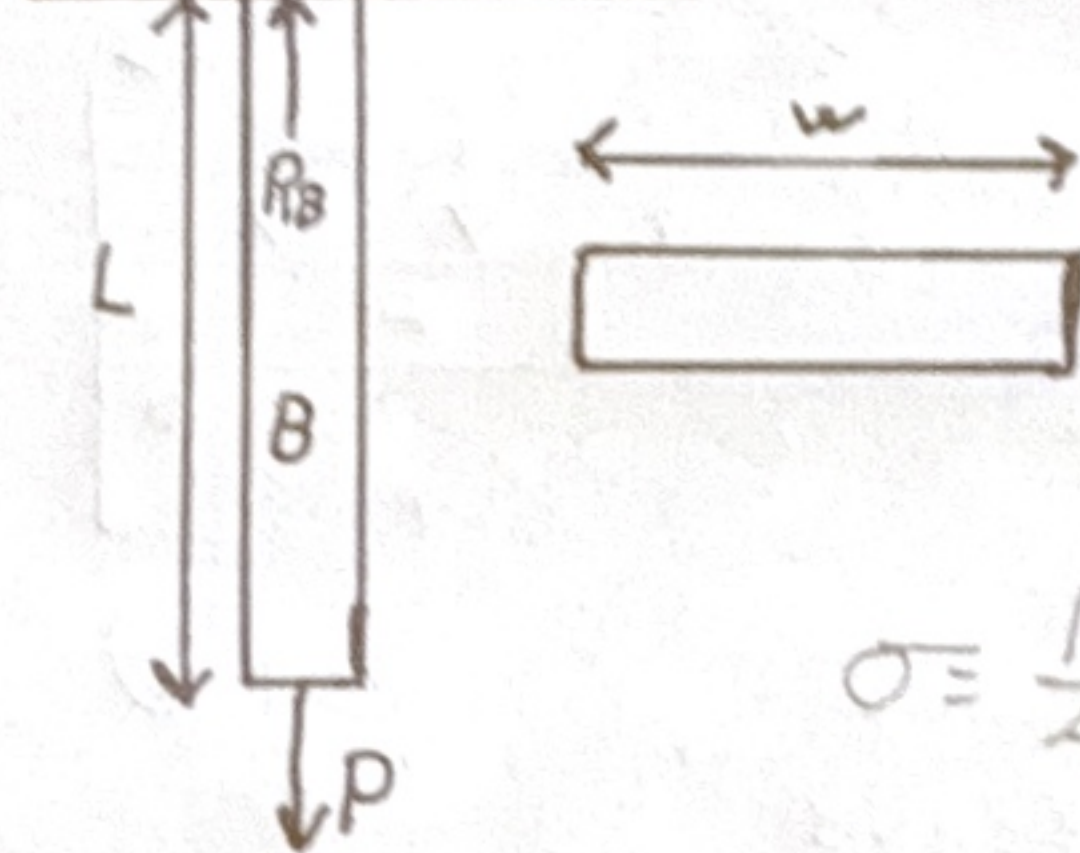
$$r = \sqrt[4]{\frac{4I}{\pi}} = \sqrt[4]{\frac{4(0.003997)}{\pi}} = 0.2671 \text{ in} \Rightarrow d = 0.534$$

Feature B

Known:

$$P = 1600 \text{ lbf}, w = 0.935 \text{ in}, h = 4, S_y = 39885 \text{ psi}, E = 10007604, \delta_{\max} = 0.005 \text{ in}$$

$$L = 1.5 \text{ in}$$



$$\sum F_y = 0 \quad R_B - P = 0 \Rightarrow R_B = 1600 \text{ lbf}$$

$$\sigma = \frac{P}{A} = \frac{S_y}{n} \Rightarrow A_{\text{req}} = \frac{nP}{S_y} = \frac{4(1600)}{39885} = 0.1605 \text{ in}^2$$

$$t = \frac{0.1605}{0.935} = \boxed{0.172 \text{ in}}$$

$$\delta = \frac{PL}{AE}$$

$$L_{\max} = \frac{\delta_{\max} AE}{P} = \frac{0.005(0.1605)(10007604)}{1600} = 5.02 \text{ in}$$

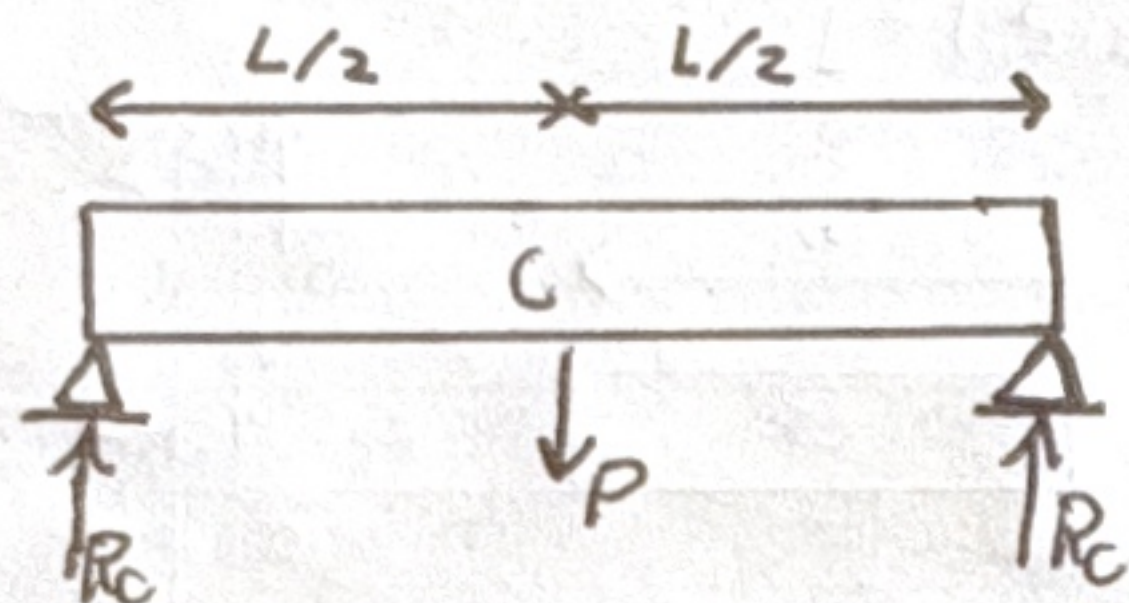
$$A_{\text{req}} = \frac{PL}{E\delta_{\max}} = \frac{1600(1.5)}{10007604(0.005)} = 0.0480 \text{ in}^2$$

$$t = \frac{A_{\text{req}}}{w} = \frac{0.0480}{0.935} = 0.051 \text{ in}$$

Feature C

$$P = R_{\theta} = 1600 \text{ lbf}, L_c = C = 1.5 \text{ in}, w = 0.935 \text{ in}, n = 4$$

$$S_y = 39885 \text{ psi}, E = 10007604 \text{ psi}, \delta_{\max} = 0.005 \text{ in}$$



$$\sum M_{\text{left}} = 0, R_{\text{right}} L - P \left(\frac{L}{2} \right) = 0$$

$$\rightarrow R_{\text{right}} = \frac{P}{2} = 800 \text{ lbf}$$

$$\sum F_y = 0, R_{\text{left}} + R_{\text{right}} - P = 0$$

$$\rightarrow R_{\text{left}} = 800 \text{ lbf}$$

$$M_{\max} = \frac{PL}{4} = \frac{1600(1.5)}{4} = 600 \text{ in} \cdot \text{lbf}$$

$$\sigma = \frac{M_{\max}}{Z} = \frac{S_y}{n} \rightarrow Z = \frac{n M_{\max}}{S_y} = \frac{4(600)}{39885} = 0.0602 \text{ in}^3$$

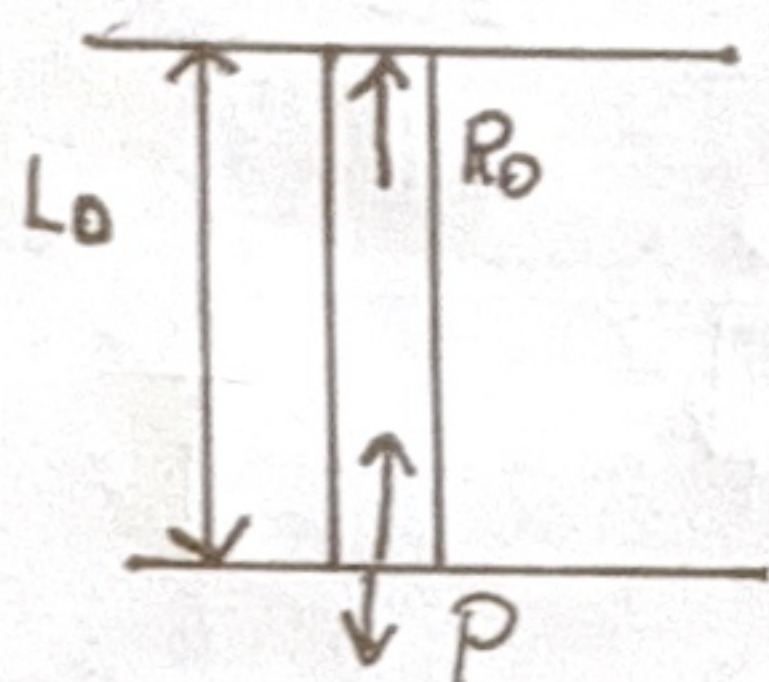
$$Z = \frac{wt_c^2}{6} \rightarrow t_c = \sqrt{\frac{6Z}{w}} = \sqrt{\frac{6(0.0602)}{0.935}} = \boxed{0.602 \text{ in}}$$

$$\delta = \frac{PL^3}{48EI} \rightarrow I = \frac{PL^3}{48E\delta_{\max}} = \frac{1600(1.5)^3}{48(10007604)(0.005)} = 0.002248 \text{ in}^4$$

$$I = \frac{wt_c^3}{12} \rightarrow t_c = \sqrt[3]{\frac{12I}{w}} = \sqrt[3]{\frac{12(0.002248)}{0.935}} = 0.307 \text{ in}$$

Feature D

$$P = R_c = 800 \text{ lbf}, L_D = D = 1 \text{ in}, w = 0.935 \text{ in}, n = 4, \\ S_y = 39885 \text{ psi}, E = 10007604 \text{ psi}, \delta_{\max} = 0.005 \text{ in}$$



$$\sum F_y = 0, R_D - P = 0 \Rightarrow R_D = 800 \text{ lbf}$$

$$\sigma = \frac{P}{A} = \frac{S_y}{n} \Rightarrow A_{\text{req}} = \frac{nP}{S_y} \quad t_d = \frac{A_{\text{req}}}{w}$$

$$A_{\text{req}} = \frac{4(800)}{39885} = 0.0802 \text{ in}^2$$

$$t_d = \frac{0.0802}{0.935} = \boxed{0.086 \text{ in}}$$

$$\delta = \frac{PL_D}{AE} \Rightarrow A_{\text{req}} = \frac{PL_D}{E \delta_{\max}}$$

$$t_d = \frac{A_{\text{req}}}{w}$$

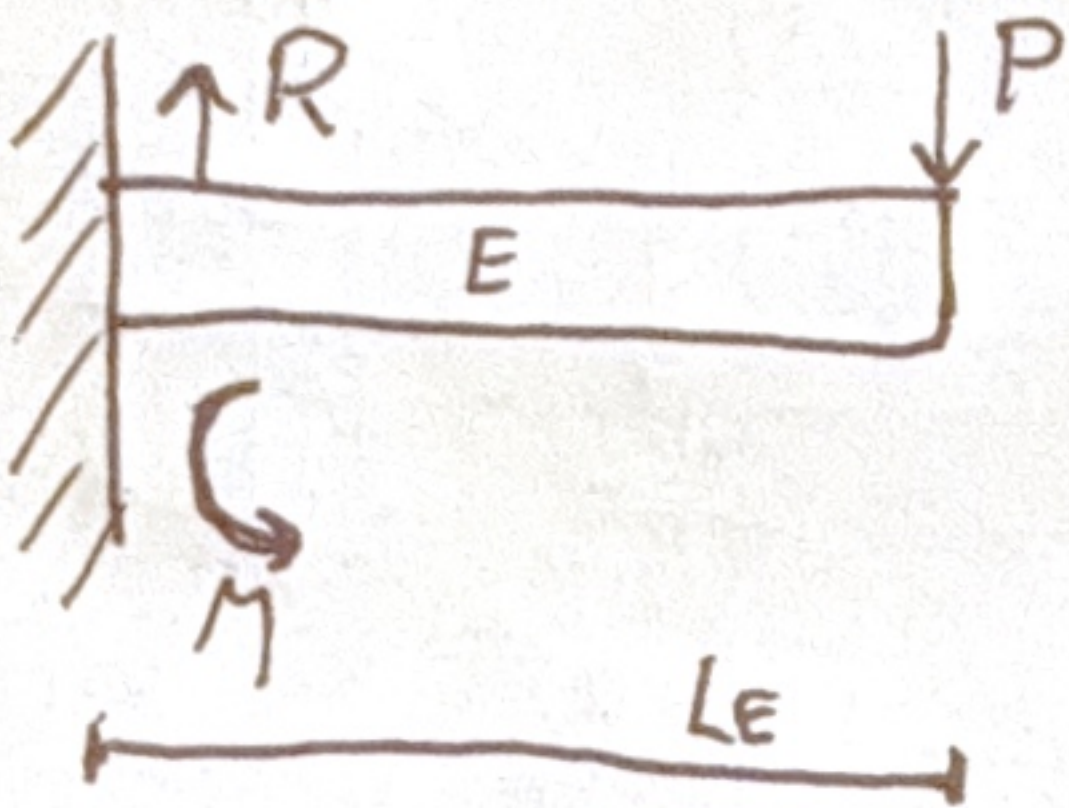
$$A_{\text{req}} = \frac{800(1)}{10007604(0.005)} = 0.016 \text{ in}^2$$

$$t_d = \frac{0.016}{0.935} = 0.017 \text{ in}$$

Feature E

$$P = R_D = 800 \text{ lbf}, \quad L_E = E = 0.5 \text{ in}, \quad w = 0.935, \quad n = 4$$

$$S_y = 39885, \quad E = 10007604 \text{ psi}, \quad \delta_{\max} = 0.005 \text{ in}$$



$$\sum F_y = 0 \quad R - P = 0 \rightarrow R = 800 \text{ lbf}$$

$$\sum M_{\text{fixed end}} = 0, \quad M - PL_E = 0 \rightarrow M = 800(0.5) = 400$$

$$\sigma = \frac{M}{z} = \frac{S_y}{n} \rightarrow z = \frac{nM}{S_y} = \frac{4(400)}{39885} = 0.0401 \text{ in}^3$$

$$z = \frac{wt_e^2}{6} \rightarrow t_e = \sqrt{\frac{6z}{w}} = \sqrt{\frac{6(0.0401)}{0.935}} = \boxed{0.507 \text{ in}}$$

$$\delta = \frac{PL_E^3}{3EI} \rightarrow I = \frac{PL_E^3}{3E\delta_{\max}} = \frac{800(0.5)^3}{3(10007604)(0.005)} = 0.000666 \text{ in}^4$$

$$t_e = \sqrt[3]{\frac{12I}{w}} = \sqrt[3]{\frac{12(0.000666)}{0.935}} = 0.204 \text{ in}$$